



ScienceDirect®

Reactive and Functional Polymers

Date: December 2025

[Show more](#) ▼Research article Full text access [Get rights and content](#) ↗

A switchable wettability coating based on KH550-SiO₂-DTMS nanocomposite particles with controllable oil-water separation performance

Chenyu Cai, Shuangshuang Liang, Tianxiang Li, Luyao Cong, An Fan, Hong Hao   [Show more](#) ▼

View PDF

[Download full issue](#)[Cite](#)[Add to Mendeley](#)[Share](#)[10.1016/j.reactfunctpolym.2025.106498](#) ↗

Article

Recommended articles

Cited by (1)

Metrics

Outline

[Highlights](#)[Abstract](#)[Graphical abstract](#)[Keywords](#)[1. Introduction](#)[2. Materials and methods](#)[3. Results and discussion](#)[4. Conclusions](#)

Highlights

- Superhydrophobic (>150°, pH 7) to superhydrophilic (0°, 20 s, pH 1), recovering superhydrophobicity (163°, pH 13).
- Maintains superhydrophobicity in various salt solutions (NaCl, KCl, KNO₃) and after ultrasonic washing.
- Achieves >98.5% efficiency for high-density oils and >96% for low-density oil-water mixtures.
- Demonstrates reliable performance for intelligent, controllable oil-water separation applications.

Abstract

Intelligently responsive super-wetting coatings have garnered significant attention for their ability to modulate wettability in response to environmental stimuli, enabling efficient oil/water separation. This study presents a novel pH-responsive super-wetting coating fabricated via a simple, cost-effective sol-gel and dip-coating approach on cotton fabric. The coating combines low-surface-energy dodecyl trimethoxysilane (DTMS) with pH-sensitive 3-aminopropyl triethoxysilane (KH550) modified SiO₂ nanoparticles. The coating's structure, morphology, and properties were characterized using Fourier transform infrared (FTIR), scanning electron microscopy (SEM), thermogravimetric analysis (TGA), X-ray photoelectron spectroscopy (XPS), and water contact angle measurements (WCA). Compared to existing pH-responsive systems, this coating exhibits superior switching speed: it transitions rapidly from superhydrophobicity (WCA > 150° at pH 7) to superhydrophilicity (WCA decreases from 50.24° to 0° within 20s at pH 1) and reversibly back to superhydrophobicity (WCA = 163° at pH 13). The coating demonstrates excellent stability in various salt solutions (NaCl, KCl, KNO₃) and under ultrasonic washing, ensuring robustness in complex aqueous environments. Leveraging this reversible wettability switching, the coating achieves high separation efficiencies for both high-density (98.5%) and low-density (96%) oil/water mixtures. Critically, this fabrication strategy eliminates the need for expensive, environmentally hazardous heavy metals (e.g., Ag, Au, Cu) or complex polymerization processes commonly employed in other smart separation materials. The combination of rapid switching, environmental and economic friendliness, salt resistance, durability, and high separation performance underscores the significant potential of this coating for diverse practical oil/water separation applications.

Graphical abstract

CRediT authorship contribution statement

Declaration of competing interest

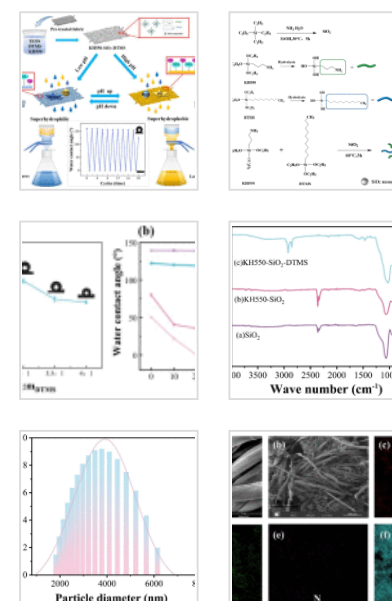
Acknowledgment

Data availability

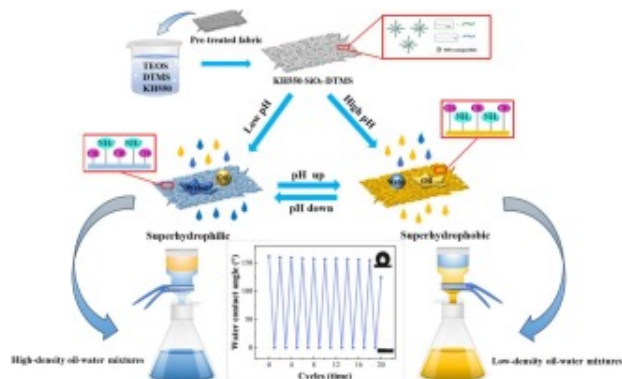
References

Show full outline 

Figures (17)



Show 11 more figures 



Download: [Download high-res image \(123KB\)](#)

Download: [Download full-size image](#)

Keywords

pH-responsive; Oil/water separation; Smart material; Switchable wettability

[< Previous article in this issue](#)

[Next article in this issue >](#)

1. Introduction

In recent years, oil spills had significant impacts on global marine and groundwater ecosystems, as well as on human health [1,2]. Consequently, there is an urgent need to develop effective strategies for oil spill remediation. Traditional oil/water separation techniques, including biodegradation, chemical coagulation, combustion, and gravity precipitation, exhibit inherent limitations. The special infiltration materials have become a simple and effective method for oil/water separation due to their special infiltration, high separation efficiency, and low cost [[3], [4], [5]]. These materials are generally categorized into three types: “oil-removing” membranes possessing superhydrophobic and superoleophilic surfaces; “water-removing” membranes characterized by

superhydrophilicity and underwater superoleophobicity [6]; smart controllable separation materials capable of switching their surface wettability in response to external stimuli [7]. Oil-removing materials offer significant convenience and are highly desirable for continuous oil/water separation systems. However, they are less effective for separating light oil/water mixtures (oil density < water density). Furthermore, these membranes are prone to oil fouling in air, resulting in the loss of their underwater superoleophobicity [5,8]. Smart controllable separation materials provide a promising solution to these challenges by dynamically adjusting surface wettability, enabling reversible switching between superhydrophobic/superoleophilic and superhydrophilic/underwater superoleophobic states [9]. These materials not only facilitate precise separation of both light oil/water (oil density < water density) and heavy oil/water (oil density > water density) mixtures, but also enhance separation rates, simplify equipment design, reduce energy consumption, and significantly improve the efficiency of oil-contaminated water treatment, thereby addressing diverse separation requirements [[10], [11], [12]].

Smart controllable separation materials can dynamically alter a material's surface wettability through modification with stimulus-responsive elements. These stimuli include changes in pH [13], temperature [14,15], light [16], gas [16], ion [17,18], solvent [19], voltage [20,21], magnetic response [22]. Chen et al. successfully fabricated an intelligent photoresponsive polypropylene (PP) membrane by grafting azobenzene groups onto SiO₂-roughened PP substrates. This photosensitive membrane enabled controllable oil/water separation under alternating UV/visible light irradiation [23]. In a separate study, Li et al. employed the SI-eATRP method to graft thermally responsive PNIPAM polymer brushes, along with non-responsive P(NP-co-AM) and P(NP-co-GM) copolymer brushes, onto the surface of a PES membrane. They further investigated the mechanism by which temperature influences the membrane's separation properties. Similarly, Zhang et al. grafted polydimethylsiloxane (PDMS) onto Fe₃O₄ particles via heat treatment to produce superhydrophobic Fe₃O₄-PDMS powder (denoted as Fe₃O₄-PDMS). This material demonstrated high oil/water separation efficiency, excellent magnetic responsiveness,

and enabled rapid oil absorption with magnetic-guided recovery [22]. Nevertheless, stimuli such as light irradiation, magnetic fields, gas exposure, and electrical current necessitate specialized equipment and stringent operational conditions, complicating the wettability switching process. In contrast, pH-responsive wettability regulation offers a simpler and more cost-effective approach, where surface properties can be readily modulated through acid/base solutions.

Research on pH-responsive wettability surfaces has gained significant prominence in oil/water separation applications. Liu et al. reported a pH-responsive coating by dipping a copolymer [PDMA-co-PTMSPMA-co-PDEAEMA] / silica mixture onto various substrates [8]. However, pH-responsive copolymer coatings typically exhibit delayed responses, often requiring several minutes to achieve activation. To address this limitation, Guo et al. developed a reversible pH-responsive smart stainless-steel mesh (SSM). This was achieved by utilizing a polydopamine adhesion layer to anchor silver (Ag) particles, followed by surface modification with 1-decanethiol [HS(CH₂)₉CH₃] and 11-mercaptopundecanoic acid [HS(CH₂)₁₀COOH] [24]. However, the preparation process employs heavy metals such as gold (Au), silver (Ag), and copper (Cu), raising significant concerns regarding both high material costs and potential heavy metal pollution. Liu et al. copolymerized styrene (a hydrophobic monomer) with 2-(dimethylamino) ethyl methacrylate (DMAEMA), which exhibits pH-dependent structural responses and degradation resistance [25]. However, this membrane undergoes mechanical weakening upon repeated pH exposure and exhibits slow structural recovery. Consequently, developing cost-effective, environmentally benign methods to fabricate pH-responsive smart surfaces with rapid wettability switching is imperative.

In this study, a coating exhibiting pH-reversible wettability was fabricated on cotton fabric substrates using sol-gel and dip-coating techniques. Surface roughness was imparted by SiO₂ particles generated via the sol-gel hydrolysis of tetraethyl orthosilicate (TEOS). To achieve low surface energy and pH responsiveness, dodecyl trimethoxysilane (DTMS), a long-chain alkylsilane, and 3-aminopropyl triethoxysilane (KH550), containing pH-sensitive amino groups, were subsequently incorporated. The pH-reversible

wettability arises from the amino groups serving as the pH-responsive moiety, while the long-chain alkylsilane reduces the surface energy.

2. Materials and methods

2.1. Materials

Tetraethyl orthosilicate (TEOS), sodium chloride (NaCl), potassium nitrate (KNO₃), and potassium chloride (KCl) were purchased from Tianjin Damao Chemical Reagent Factory. Ammonium hydroxide (NH₄OH), hydrochloric acid (HCl), sodium hydroxide (NaOH), n-hexane, petroleum ether, and toluene were obtained from Aladdin Reagent (Shanghai). Dichloromethane, chloroform, and carbon tetrachloride were supplied by Yonghua Chemical Co. Dodecyltrimethoxysilane (DTMS, 95%) and 3-aminopropyltriethoxysilane (KH550, 99%) were procured from McLean Biochemical Technology Co. (Shanghai).

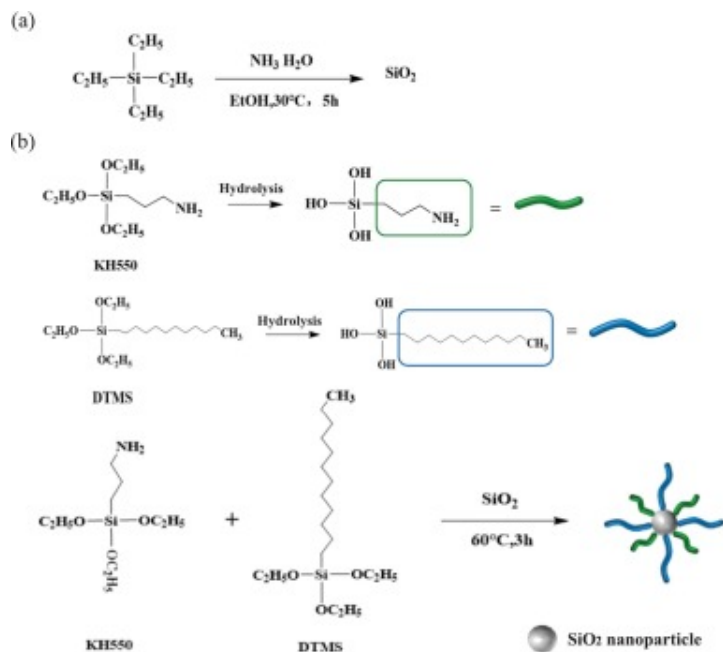
2.2. Experimental work

2.2.1. Pre-treatment of raw substrates

Raw cotton fabrics were cut into 2×2 cm pieces and immersed in a mixed solvent system (anhydrous ethanol/acetone/deionized water=1:1:2 v/v). Ultrasonic cleaning was performed for 20 min, followed by three sequential rinses with alternating ethanol and deionized water. The pre-treated pristine cotton fabrics were subsequently dried at 60 °C in an oven for 2 h.

2.2.2. Preparation of silicon dioxide suspension

In a round-bottom flask, anhydrous ethanol (15 mL) and ammonia water (10 mL) were mixed. Tetraethyl orthosilicate (TEOS) (5 mL) was then added dropwise to the solution under continuous stirring. The mixture underwent stirring at 30 °C for 5 h to yield a homogeneous white SiO₂ suspension via the sol-gel process. The schematic of TEOS hydrolysis is presented in [Fig. 1a](#).



[Download: Download high-res image \(109KB\)](#)

[Download: Download full-size image](#)

Fig. 1. (a) Schematic diagram of TEOS hydrolysis. (b) Schematic diagram of KH550- SiO₂-DTMS synthesis.

2.2.3. Preparation of the pH-responsive coating suspension

SiO₂ suspension (12 mL), KH550 (12 mL) and DTMS (0.4 mL) were added to a three-necked flask and ultrasonically dispersed for 30 min. The resulting mixture was subsequently stirred at 60 °C for 3 h, yielding a KH550-SiO₂-DTMS suspension. The preparation process is schematically illustrated in Fig. 1b.

2.2.4. Preparation of the pH-responsive wettable coating

The cotton fabric specimens were immersed in the KH550-SiO₂-DTMS suspension for 10 min. Three controlled immersion-withdrawal cycles were performed at uniform speed

to generate a uniform KH550-SiO₂-DTMS coating.

2.3. Characterization

The chemical structure of the coating was characterized by Fourier Transform Infrared (FTIR) spectroscopy (Bruker Vertex70, Germany) over a wavenumber range of 4000–400 cm⁻¹. Static contact angles (CAs) were measured at three distinct locations on the coating surface using a DCAT21 goniometer (Dataphysics, Germany) via the sessile drop method, and averaged values reported. Surface elemental composition and morphology were analyzed by energy-dispersive X-ray spectroscopy (EDX) coupled with scanning electron microscopy (SEM; Carl Zeiss SIGMA field, Germany). Chemical states and elemental compositions were analyzed by X-ray photoelectron spectroscopy (XPS) using a Thermo Scientific ESCALAB Xi+ spectrometer (Thermo Fisher, USA). Particle size distributions were determined with a Malvern Nano-ZS ZEN3600 laser particle sizer (England), with triplicate measurements per sample. Thermogravimetric analysis (TGA) was performed on a Netzsch TG209 instrument (Germany) under N₂ flow (60 mL/min), with temperature ramping from ambient to 900 °C at 10 °C/min.

2.4. Reversible cycle test

The initial KH550-SiO₂-DTMS coating was designated as Coating 1, and its water contact angle (WCA) was measured. Coating 1 was then immersed in an acidic solution (pH 1), oven-dried, and relabeled as Coating 2, followed by WCA measurement. Subsequently, Coating 2 was immersed in an alkaline solution (pH 13), oven-dried, and designated as Coating 3, with WCA remeasured. This acid-base cycling procedure was repeated for three complete cycles to evaluate the coating's reversible wettability switching capability.

2.5. Salt resistance test

Three replicate samples of the KH550-SiO₂-DTMS coating were separately immersed in 1 M NaCl, 1 M KCl, and 1 M KNO₃ solutions for 12 h. WCA were measured every 2 h to evaluate the coating's salt tolerance based on contact angle stability.

2.6. Mechanical stability and ultrasonic durability tests

2.6.1. Mechanical stability test

Abrasion resistance tests were conducted to evaluate the mechanical durability of the KH550- SiO₂-DTMS coating. The coated fabric sample was placed face-down on 800-grit sandpaper and subjected to a 100g load. The sample was then dragged horizontally over a distance of 20cm, constituting one abrasion cycle. This process was repeated for 20cycles. The change in WCA was measured after abrasion.

2.6.2. Ultrasonic durability tests

A homogeneous mixture of ethanol and deionized water (1:1 v/v) was prepared. The KH550- SiO₂-DTMS coating was then immersed in this solution and subjected to ultrasonic cleaning for 30min. Following drying, the WCA was measured. This cycle was repeated to evaluate the ultrasonic cleaning resistance of the KH550- SiO₂-DTMS coating.

2.7. Oil/water separation performance test

2.7.1. High-density oil/water separation process

Original cotton fabric was cut to the appropriate size to fit the filter apertures and then impregnated with the prepared KH550-SiO₂-DTMS suspension to yield the superhydrophobic/superoleophilic KH550- SiO₂-DTMS coating. The coated fabric was secured within the filter housing. Subsequently, various distinctly colored oil/water mixtures were poured onto the upper layer and allowed to stand. The separation performance for the different oil/water mixtures was evaluated by monitoring the gravity-driven filtration of oil and measuring the residual water levels. The oil–water separation efficiency was calculated using the following equation:

$$R = M / M_0 \times 100\%$$

Where M_0 and M are the original oil weight and the collected oil weight, respectively. To ensure the reliability of the data, all experiments were carried out 3 times.

2.7.2. Low-density oil/water separation process

The superhydrophobic/superoleophilic KH550-SiO₂-DTMS coating was immersed in a strongly acidic HCl solution (pH1) to induce a wettability transition (serving as the coating pre-wetting step). The resultant KH550- SiO₂-DTMS coating, now exhibiting superhydrophilic and underwater superoleophobic properties, was secured within the filter apertures. Distinctly colored oil/water mixtures were then poured onto the coated surface and allowed to stand. The separation performance for the different oil/water systems was quantified by monitoring gravity-driven filtration and measuring the residual oil content.

2.7.3. Cyclic reusability test for oil/water separation

To investigate the cyclability (reusability) of the KH550- SiO₂- DTMS and acid-treated KH550- SiO₂-DTMS coatings for cyclic reusability testing for oil/water separation, the coating was dried after each separation cycle prior to the subsequent test. The separation efficiency after 8cycles was determined to evaluate the coating durability.

2.7.4. Flux measurement

The separation flux (J) was calculated to evaluate the processing capacity of the coated fabric, denoted as equation:

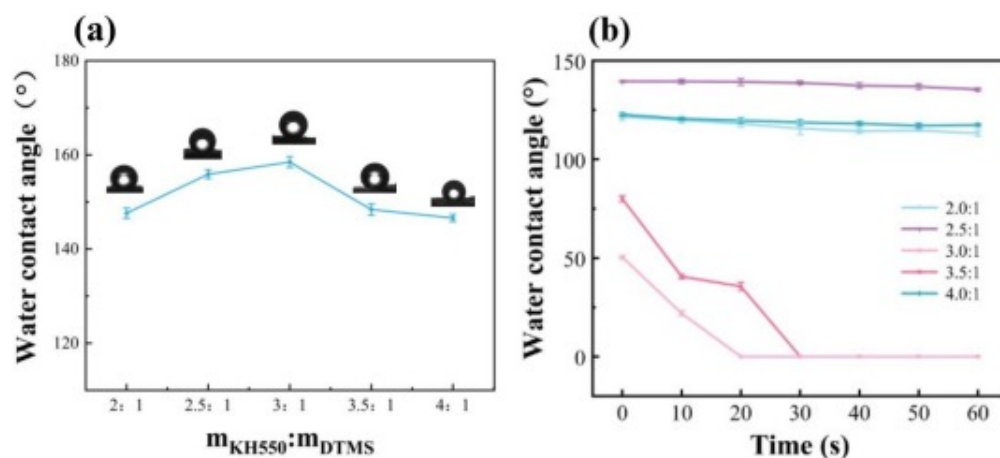
$$J = V / A \times t$$

where V (L) represents the penetration volume (oil or water), A (m²) represents the penetration area, and t (h) represents the penetration time. All flux measurements were conducted under gravity and repeated at least three times.

3. Results and discussion

3.1. The effect of KH550/DTMS on wettability and transformation properties

The KH550/DTMS mass ratio was employed as a key variable to modulate the properties of the coating. SiO₂ suspension (12 mL) was incorporated into the reaction mixture at 60°C. The wettability and transition behavior of the resulting KH550-SiO₂-DTMS coatings, prepared from suspensions with varying KH550/DTMS mass ratios, were investigated. The static WCAs of these coatings measured at pH7 are presented in Fig. 2a. The WCAs were 143.5°, 155.9°, 158.5°, 146.2°, and 136.0° for KH550/DTMS ratios of 2:1, 2.5:1, 3:1, 3.5:1, and 4:1, respectively. Notably, the oil contact angle (OCA) remained 0° for all ratios. Subsequently, coatings prepared with different KH550/DTMS mass ratios were immersed in a strongly acidic HCl solution (pH1) for 2 min. After drying, the dynamic change in WCA at pH7 was monitored, with measurements recorded every 10s (Fig. 2b). For the coating with a KH550/DTMS ratio of 3:1, the WCA decreased sharply from 50.24° (stated earlier as initial value after acid treatment and drying) to 0° within 20s. The transition time of the WCA before and after acid treatment was recorded for each coating.



Download: [Download high-res image \(81KB\)](#)

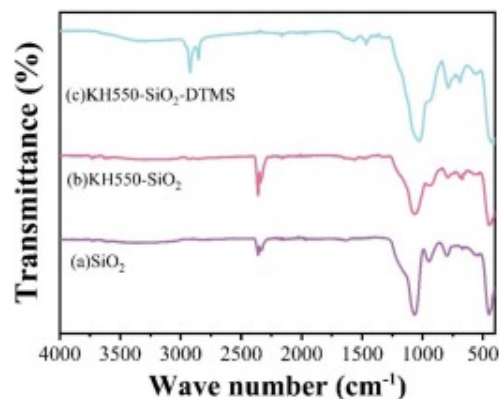
Download: [Download full-size image](#)

Fig. 2. (a) Effect of different contents of KH550 on the wettability of the coating. (b) Changes in the wettability of the coating prepared with different contents of KH550 in 1 min after treatment with the acid solution.

Research results indicate that the amino groups ($-\text{NH}_2$) in KH550 play a critical role in regulating surface wettability. Under acidic conditions: For a KH550: DTMS ratio $< 3:1$: An excess of DTMS leads to excessive aggregation of alkyl chains. This suppresses the formation of SiO₂ nano-rough structures, consequently diminishing the coating's hydrophobicity. Simultaneously, insufficient amino groups result in fewer protonation sites, leading to slower hydrophilic transition kinetics. KH550: DTMS ratio $> 3:1$: An abundance of amino groups begins to dominantly compromise the coating's hydrophobicity. Furthermore, intermolecular hydrogen bonding between amino groups can induce chain-like agglomeration, thereby disrupting the surface micro/nano composite structure.

3.2. Infrared spectral analysis

Fourier transform infrared (FTIR) spectroscopy was performed on SiO₂, KH550-SiO₂, and KH550-SiO₂-DTMS samples, with the results presented in Fig. 3. Peaks at 2852cm^{-1} and 2945cm^{-1} correspond to the asymmetric stretching vibrations of $-\text{CH}_2$ and $-\text{CH}_3$ groups, respectively. The broad absorption band spanning $950\text{--}1070\text{cm}^{-1}$ is attributed to Si-O-Si asymmetric stretching vibrations. The peak at 796cm^{-1} is assigned to the Si-C stretching region. Comparing the KH550-SiO₂ spectrum to that of SiO₂, a new peak emerges at 1570cm^{-1} . This band is characteristic of the N-H bending (scissoring) vibration of primary amines ($-\text{NH}_2$). The appearance and increased intensity of this peak confirm the successful grafting of KH550 onto the SiO₂ surface.

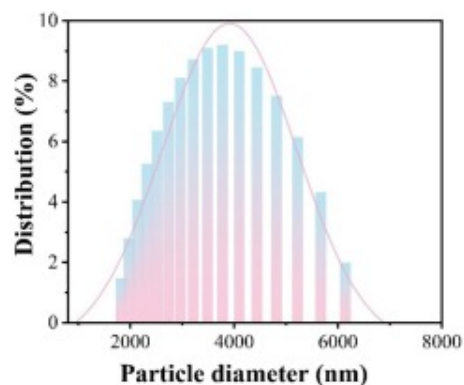


[Download: Download high-res image \(88KB\)](#)

[Download: Download full-size image](#)

Fig. 3. FT-IR spectra of (a) SiO₂. (b) KH550-SiO₂. (c) KH550-SiO₂-DTMS.

The particle size distribution of the composite suspension was characterized using laser diffraction particle size analysis. As shown in Fig. 4, the mean droplet diameter of the KH550-SiO₂-DTMS suspension was measured to be approximately 2964 nm. Furthermore, the obtained film presented a uniform surface morphology, and no whitening phenomenon was observed, indicating that the coating had good dispersion on a macroscopic scale and no severe particle agglomeration occurred.



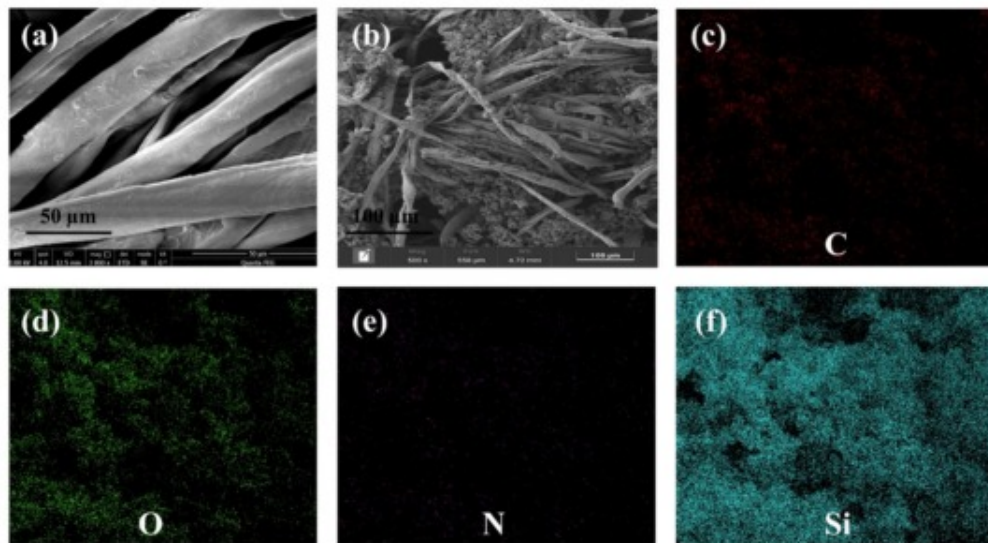
[Download: Download high-res image \(62KB\)](#)

[Download: Download full-size image](#)

Fig. 4. Distribution of SiO₂ particle size under optimal preparation conditions.

3.3. The surface morphology analysis

The morphologies of the original cotton fabric and the KH550-SiO₂-DTMS coating were characterized by (SEM), as shown in Fig. 5. Fig. 5a reveals that the surface of the original cotton fabric exhibits a smooth and uniform ribbon-like morphology. In contrast, the KH550-SiO₂-DTMS coating surface (Fig. 5b) displays a distinct microstructure comprising well-dispersed nanoparticles with an average diameter of approximately 400nm, large irregular microscale compact aggregates (~5μm), and micropores throughout. The image clearly shows that the nanocomposite particles are uniformly grafted onto the surface of the cotton fabric. The robust adhesion of the KH550-SiO₂-DTMS coating to the cotton fabric is attributed to a synergistic combination of covalent bonding and physical anchoring. During the dip-coating process, the hydrolyzed silanol groups (–Si–OH) derived from KH550 and DTMS undergo a condensation reaction with the hydroxyl groups (–OH) on the cellulose fibers, forming stable Si–O–Cellulose covalent bonds. This chemical grafting is further enhanced by physical interlocking: the micro-nano hierarchical structures composed of SiO₂ aggregates and the silane network penetrate and anchor mechanically within the porous fibrous matrix of the cotton. This structure collectively forms a highly roughened surface conducive to super-infiltration. Elemental mapping analysis of the KH550-SiO₂-DTMS coating surface (Fig. 5c–f) confirmed the homogeneous distribution of C, O, Si, and N elements. This homogeneous elemental distribution indicates that the coating surface is predominantly composed of a mixture of SiO₂, KH550, and DTMS.



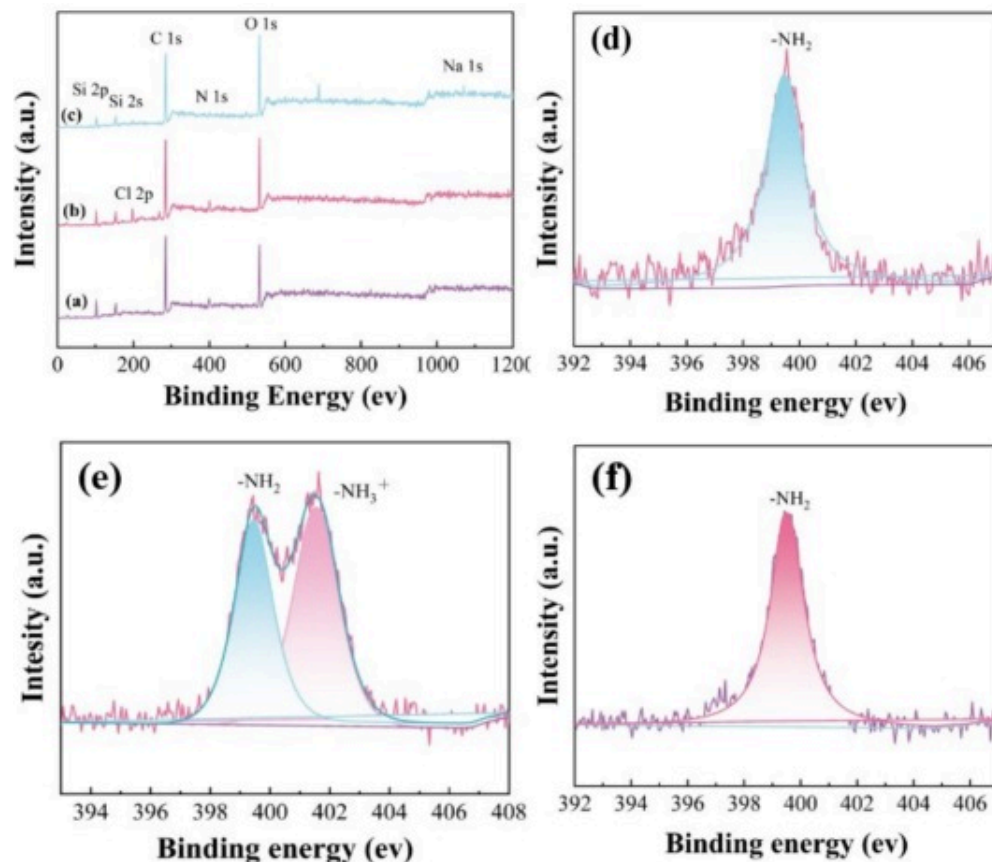
Download: [Download high-res image \(196KB\)](#)

Download: [Download full-size image](#)

Fig. 5. SEM plots of (a) original cotton fabric, (b) KH550-SiO₂-DTMS coating surface, EDS plots of different elements, (c) C, (d) O, (e) N, (f) Si.

3.4. XPS analysis

XPS analysis was performed on the pristine KH550-SiO₂-DTMS coating, as well as acid-treated (pH1 HCl solution) and alkali-treated (pH13 NaOH solution) samples; the results are presented in Fig. 6a. The pristine KH550-SiO₂-DTMS coating exhibited characteristic peaks corresponding to C 1s (285.17 eV), Si 2s (153.68 eV), Si 2p (102.76 eV), N 1s (401.37 eV), and O 1s (532.60 eV). Following treatment with HCl solution (pH1), a new Cl 2p peak emerged at 199.51 eV on the coating surface (Fig. 6b). Similarly, treatment with NaOH solution (pH13) resulted in the appearance of a new Na 1s peak at 1071.42 eV.



[Download: Download high-res image \(253KB\)](#)

[Download: Download full-size image](#)

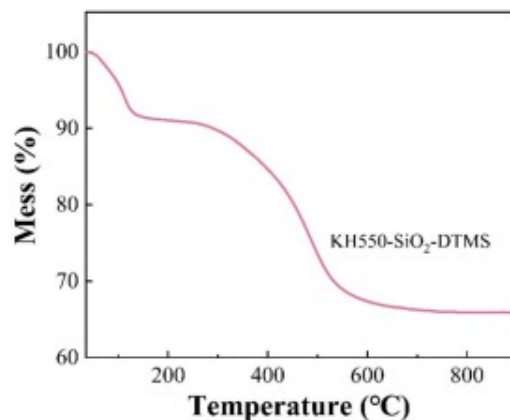
Fig. 6. XPS full spectra of (a) pristine KH550-SiO₂-DTMS; (b) acid-treated KH550-SiO₂-DTMS; (c) alkali-treated KH550-SiO₂-DTMS; Split-peak spectra of N 1s: (d) pristine KH550-SiO₂-DTMS; (e) acid-treated KH550-SiO₂-DTMS; (f) alkali-treated KH550-SiO₂-DTMS.

The high-resolution N 1s spectrum of the pristine KH550-SiO₂-DTMS coating (Fig. 6b) displayed a single peak at 399.48 eV, assigned to the -NH₂ groups originating from the KH550 silane. The inherent super-hydrophobicity of the pristine surface is attributed to the long alkyl chains of DTMS. After acid treatment, deconvolution of the high-resolution

N 1s spectrum (Fig. 6c) revealed two distinct peaks: the original -NH₂ peak at 399.48 eV and a new peak at 401.53 eV, characteristic of -NH₃⁺. This indicates protonation of the -NH₂ groups to form -NH₃⁺, which exhibits strong affinity for water, thereby switching the surface wettability to super-hydrophilic. Subsequent alkali treatment restored the N 1s spectrum to a single peak at 399.48 eV (Fig. 6d), consistent with deprotonation of -NH₃⁺ back to -NH₂. At this stage, the surface reverted to super-hydrophobicity, governed by the long alkyl chains of DTMS. Therefore, the reversible transformation between -NH₂ and -NH₃⁺ groups serves as the key factor driving the wettability transition.

3.5. TGA analysis

The thermal stability of the KH550-SiO₂-DTMS coating was evaluated by thermogravimetric analysis (TGA the thermogram is shown in Fig. 7. The TGA curve reveals three distinct stages of mass loss. An initial mass loss of 8.66% occurred below 150°C, attributed primarily to the evaporation of residual solvents such as ethanol. Subsequently, a significant mass loss of 25.41% was observed between 150°C and 550°C. This stage corresponds to the thermal decomposition involving the random scission of C—C, C—O, C—H, and C—N bonds within the organic components (KH550 and DTMS). The residual mass at 800°C was 65.92%, which is predominantly attributed to the silica (SiO₂) component. These TGA results corroborate the successful synthesis of KH550-SiO₂-DTMS composites, comprising well-dispersed SiO₂ nanoparticles (NPs) functionalized with surface-attached aminopropyl (from KH550) and alkyl chains (from DTMS).



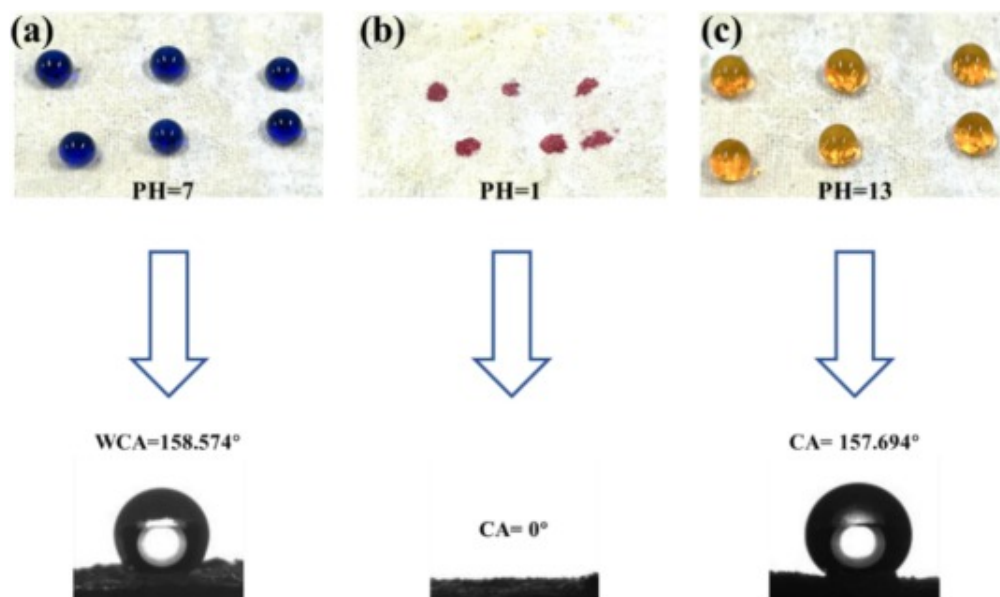
Download: [Download high-res image \(54KB\)](#)

Download: [Download full-size image](#)

Fig. 7. TG diagram of the KH550-SiO₂-DTMS.

3.6. The influence of different pH environments on the wettability of coating

In-situ wettability tests were performed on the coating. Droplets of NaOH solution (pH13, stained with methyl orange; Fig. 8c) and water (pH7, stained with methylene blue; Fig. 8a) deposited on the cotton fabric retained near-spherical shapes, demonstrating superhydrophobic behavior with WCA of 158.574° and 157.694°, respectively. In contrast, droplets of HCl solution (pH1, stained with methyl red; Fig. 8b) spread and were completely absorbed by the cotton fabric within 1 min, indicating an instantaneous switch to superhydrophilic wettability.



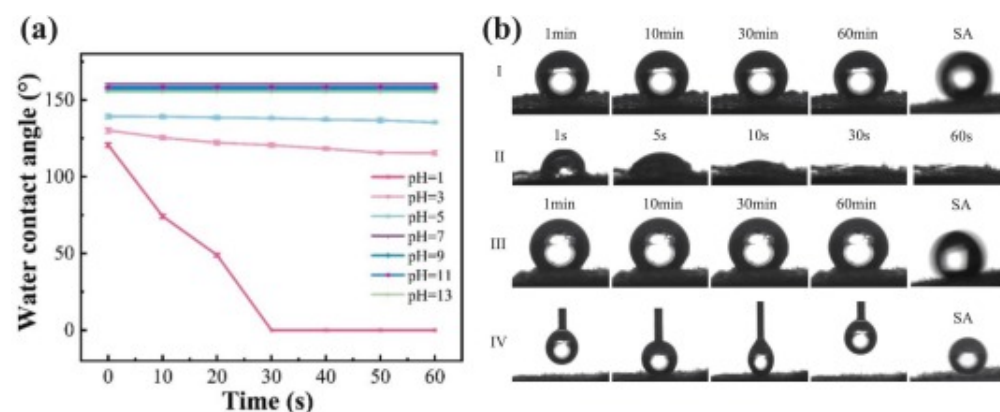
Download: [Download high-res image \(169KB\)](#)

Download: [Download full-size image](#)

Fig. 8. (a) H₂O droplet at pH7 (methylene blue staining); (b) HCl droplet at pH1 (methyl red staining); (c) NaOH droplet at pH13 (methyl orange staining). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

To evaluate the pH-responsive wettability of the KH550-SiO₂-DTMS coating, samples were immersed in solutions ranging from pH1 to pH13 (specifically pH1, 3, 5, 7, 9, 11, and 13). After drying, the WCA was measured to assess wettability changes. Furthermore, the dynamic wettability transition was monitored in situ by depositing droplets of the respective pH solutions onto the coating and recording the contact angle evolution at 10s intervals over 1 min; these results are presented in Fig. 9a. Initially, the pristine coating showed the WCA of 158.32°, which remained stable over time. Water droplets easily rolled off the surface with a sliding angle (SA) of 5.76°, demonstrating its superhydrophobicity. (Fig. 9b). After the coated cotton fabric was immersed in HCl (pH1)

for 1 min and dried, a water droplet was completely absorbed within 30s (Fig. 9bII), indicating a transition to a superhydrophilic surface. Notably, when the acid-treated fabric was further immersed in NaOH solution (pH13) for 1 min and dried, the coated fabric switched back to the superhydrophobic state and remained stable over time with an extremely SA (Fig. 9bIII). Furthermore, the acid-treated superhydrophilic surface also exhibited underwater superoleophobicity. When a 3 μ L oil droplet (e.g., dichloromethane) was pressed against the fabric surface and subsequently lifted, it showed no deformation and readily rolled off at a SA below 5°, confirming underwater superoleophobicity with ultralow oil adhesion (Fig. 9bIV).



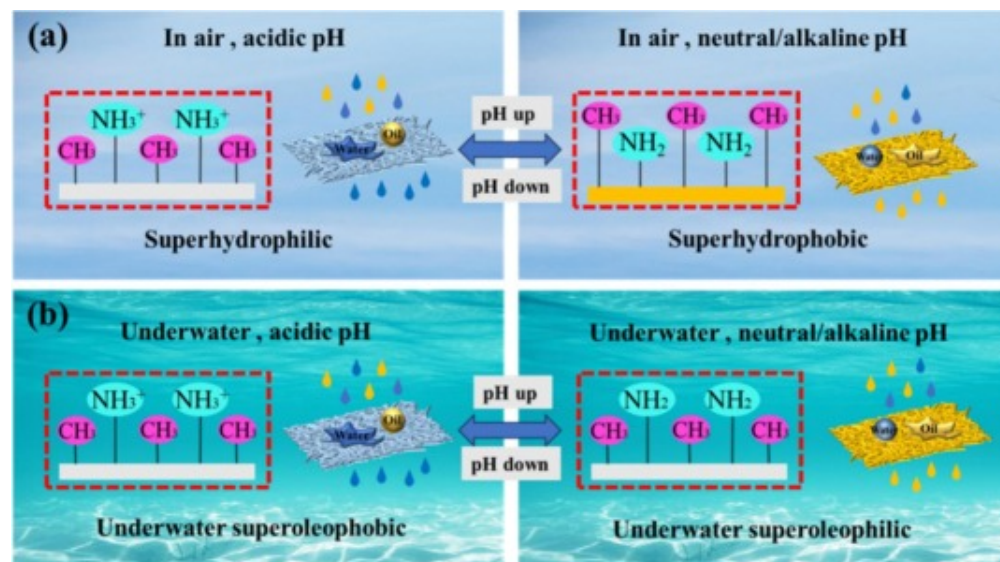
Download: [Download high-res image \(160KB\)](#)

Download: [Download full-size image](#)

Fig. 9. (a) Changes in coating wettability over 1 min after treatment with different pH solutions; (b) Characterization of coating wettability and underwater adhesion: (I) Wettability change and SA measurement of the pristine coating. (II) Wettability change and SA measurement of the coating after HCl (pH=1) treatment. (III) Wettability change and SA measurement of the coating after NaOH (pH=13) treatment. (IV) Underwater oil adhesion test and SA measurement after acid treatment.

This pH-dependent wettability transition is attributed to the protonation of surface -NH₂ groups (Fig. 10). Under acidic pH, amino groups (-NH₂) protonate into hydrophilic -NH₃⁺,

attracting water molecules that infiltrate the micro-roughness, resulting in a superhydrophilic ($\text{WCA} \approx 0^\circ$) and underwater superoleophobicity. Under alkaline conditions, the coating's wettability reverted to the superhydrophobic state, with the recovered WCA showing minimal deviation from its original value. This restoration results from the deprotonation of $-\text{NH}_3^+$ back to $-\text{NH}_2$, allowing the long hydrophobic alkyl chains of DTMS to dominate the surface properties. A maximum recovered contact angle of 163° was achieved at pH 13.



Download: [Download high-res image \(202KB\)](#)

Download: [Download full-size image](#)

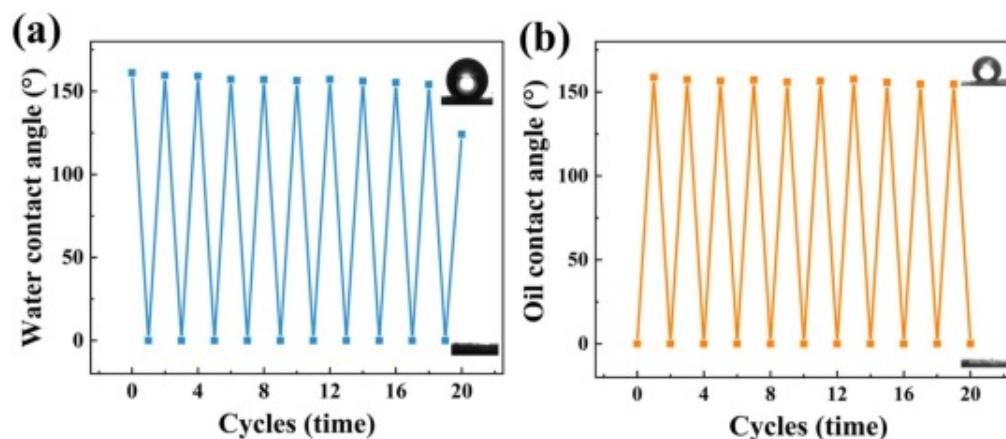
Fig. 10. Mechanism of reversible wettability transition: (a) In air, superhydrophilicity (acidic, $-\text{NH}_3^+$) switches to superhydrophobicity (neutral/alkaline, $-\text{NH}_2$ /alkyl chains). (b) Underwater, superoleophobicity under acidic condition reverses upon deprotonation.

3.7. Reversible cycle test for wettability

The coating initially exhibited superhydrophobic behavior, with a WCA of 161.14° and OCA of 0° . After immersion in an HCl solution (pH 1) for 1 min followed by drying, the WCA

decreased to 0° and OCA increased to 158.64°, indicating a transition to a Superhydrophilic/underwater superoleophobic state. Water droplets deposited on this surface spread rapidly within 3 s. Subsequently, immersion in a NaOH solution (pH 13) and drying restored the surface, enhancing the WCA to 159.44° and confirming the recovery of superhydrophobicity.

These results demonstrate the reversible wettability switching capability of the KH550-SiO₂-DTMS coating. This reversible behavior was assessed through multiple switching cycles. After nine cycles, the coating retained its superhydrophobicity with a WCA of 152.14°. However, by the tenth cycle, the WCA decreased to 124.12°, indicating a transition to a hydrophobic state. This decline is attributed to the partial abrasion of the long alkyl chains within the coating, as illustrated in Fig. 11. Despite this gradual degradation, the coating demonstrated robust reusability over multiple cycles.



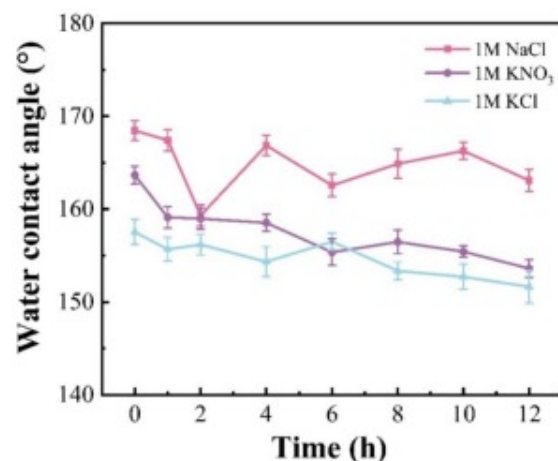
[Download: Download high-res image \(130KB\)](#)

[Download: Download full-size image](#)

Fig. 11. Changes in wettability of KH550-SiO₂-DTMS coating treated with alternating acid and alkaline solutions; (a) WCA; (b) OCA.

3.8. Salt tolerance test

To evaluate the salt resistance of the KH550-SiO₂-DTMS coating, three samples were immersed in different salt solutions (1 M NaCl, 1 M KCl, 1 M KNO₃), as shown in Fig. 12. Prior to immersion, the WCAs were 168.16° (NaCl), 157.43° (KCl), and 163.24° (KNO₃). After soaking in their respective salt solutions for 12h, the WCAs decreased slightly to 163.17° (NaCl), 151.22° (KCl), and 153.27° (KNO₃), corresponding to reductions of 4.99°, 6.21°, and 9.98°, respectively. Despite these minor reductions, all coatings retained their superhydrophobicity, demonstrating the significant salt resistance of the KH550-SiO₂-DTMS coating.



[Download: Download high-res image \(76KB\)](#)

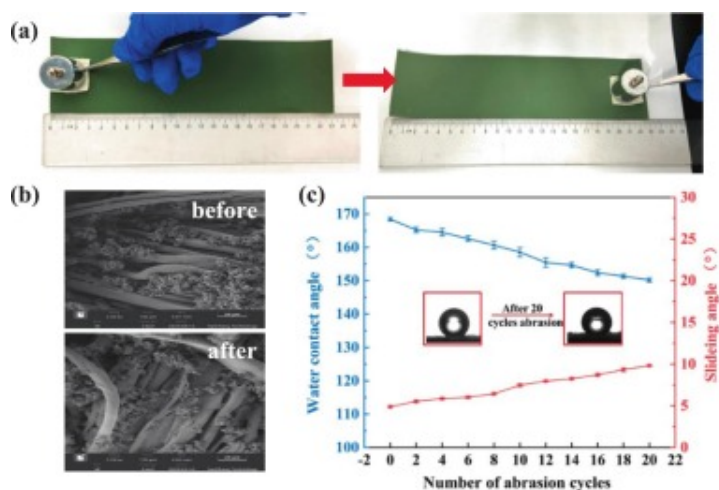
[Download: Download full-size image](#)

Fig. 12. Changes in coating wettability over 12h after treatment with different salt solutions.

3.9. Mechanical stability and ultrasonic durability analysis

The mechanical stability of the KH550-SiO₂-DTMS coating was evaluated via abrasion resistance tests (Fig. 13a). After 20 abrasion cycles, the WCA decreased from 168.54° to 150.34°, while the SA increased from 4.89° to 9.84°(Fig. 13c). The coating surface continued to exhibit superhydrophobic properties. It is noteworthy that no obvious

physical damage was observed on the coating, demonstrating its good mechanical durability (Fig. 13b).

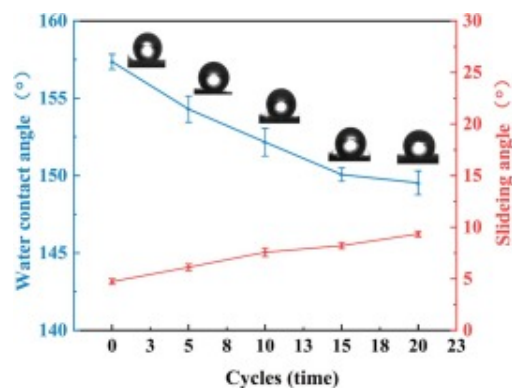


[Download: Download high-res image \(180KB\)](#)

[Download: Download full-size image](#)

Fig. 13. (a) Schematic of the abrasion test; (b) The surface of the coating before and after the abrasion test (c) Plot of the WCA and SA after abrasion cycles.

The durability of the KH550-SiO₂-DTMS coating was assessed through ultrasonic washing tests (Fig. 14). Following 20 ultrasonic washing cycles (30min per cycle), the WCA decreased from 157.14° to 150.70°, while the SA increased from 4.25° to 9.24°. Despite this reduction, the coating retained its superhydrophobicity, demonstrating robust durability.



Download: [Download high-res image \(82KB\)](#)

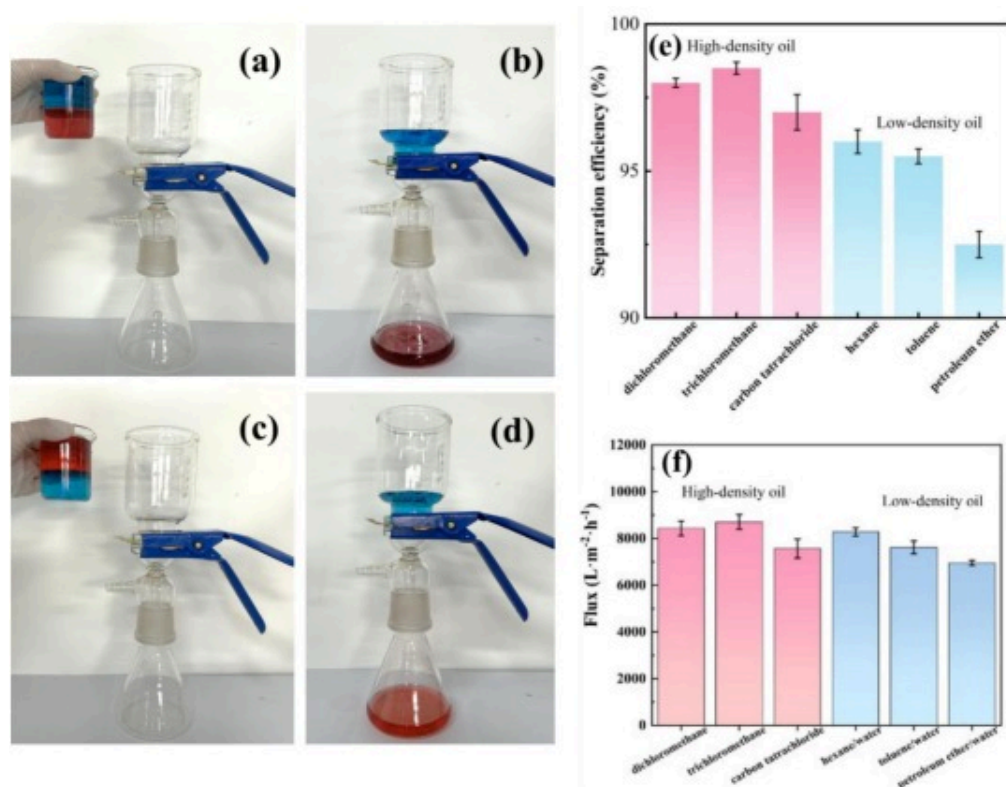
Download: [Download full-size image](#)

Fig. 14. Change in wettability after 20 ultrasonic washing cycles.

3.10. Oil/water separation performance test

3.10.1. High-density oil/water

The oil/water separation performance of the KH550-SiO₂-DTMS coated cotton was evaluated using mixtures of neutral water (stained with methylene blue) and organic solvents as the oil phase: dichloromethane, chloroform (trichloromethane), and carbon tetrachloride (stained with methyl red). In the dichloromethane/water mixture, dichloromethane rapidly permeated through the coated cotton and was collected in the lower funnel, while the water phase was effectively retained above the coating and remained stable for 5 min (Fig. 15a–b). Intelligent and selective separation of oil/water mixtures was successfully achieved by adjusting the pH values. The separation process exhibited high flux rates (Fig. 15f), reaching 8580 Lm⁻²·h⁻¹ for dichloromethane, 8875 Lm⁻²·h⁻¹ for chloroform, and 7638 Lm⁻²·h⁻¹ for carbon tetrachloride. The separation efficiencies for the three oil/water mixtures (Fig. 15e), demonstrated excellent performance: 98% for dichloromethane, 98.5% for chloroform, and 97% for carbon tetrachloride.



Download: [Download high-res image \(163KB\)](#)

Download: [Download full-size image](#)

Fig. 15. (a–d) Separation process of KH550-SiO₂-DTMS coating; (e) Separation efficiency of different types of oils; (f) Separation fluxes of types of oils.

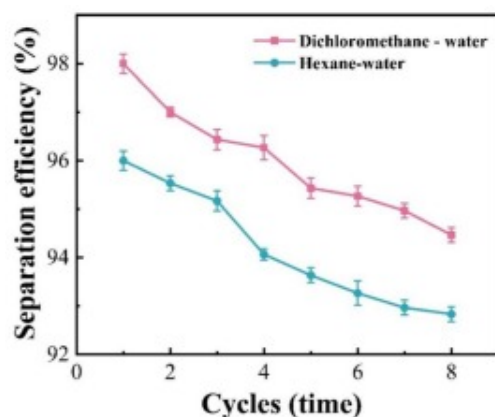
3.10.2. Low-density oil/water

After wetting the KH550-SiO₂-DTMS coating with HCl solution (pH1), oil/water separation experiments were conducted using toluene, n-hexane, and petroleum ether (all less dense than water and stained with methyl red) as the oil phase, with distilled water (pH7, stained with methylene blue) as the aqueous phase. In the n-hexane/water system (Fig. 15c–d), water rapidly permeated through the modified cotton membrane into the lower funnel, while n-hexane was effectively retained above the coating and

remained stable for 5 min. The water flux rates for low-density oil separation were measured at $8231 \text{ L} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$ for hexane/water, $7629 \text{ L} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$ for toluene/water, and $6933 \text{ L} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$ for petroleum ether/water systems. Separation efficiencies for the three mixtures (Fig. 15e) reached 96% for n-hexane, 95.5% for toluene, and 92.5% for petroleum ether.

3.10.3. Reusability analysis

The reusability test for Oil-Water Separation of the KH550-SiO₂-DTMS coating was evaluated, and the results are presented in Fig. 16. After eight separation cycles, the coating maintained a separation efficiency exceeding 94.5%, while the acid-treated KH550-SiO₂-DTMS coating retained an efficiency above 92.8%. These findings demonstrate that the coating exhibits excellent reusability and stable performance during repeated separation cycles.



[Download: Download high-res image \(61KB\)](#)

[Download: Download full-size image](#)

Fig. 16. Cyclic testing efficiency of KH550-SiO₂-DTMS coating.

4. Conclusions

This study demonstrates a straightforward and efficient approach for fabricating an intelligent coating with pH-responsive reversible wettability. A homogeneous SiO₂ suspension was synthesized via the sol-gel method using TEOS, ammonia, and ethanol. Functionalization was achieved by grafting the low-surface-energy agent DTMS and the pH-responsive aminosilane KH550 onto SiO₂ nanoparticles. Optimization of the KH550/DTMS mass ratio significantly reduced the wettability transition time to 20s. XPS analysis revealed the key switching mechanism, protonation of -NH₂ groups and deprotonation of -NH₃⁺ groups within the KH550-SiO₂-DTMS film. The functionalized coatings exhibit exceptional pH responsiveness, maintaining stable superhydrophobicity under neutral/alkaline conditions. In acidic environments, the WCA decreases markedly with pH reduction. Subsequent alkali treatment fully regenerates the superhydrophobic state. The optimal pH for wettability transition is 1, while superhydrophobicity recovery is optimal at pH 13. Furthermore, the KH550-SiO₂-DTMS coating demonstrates outstanding salt resistance and stability during ultrasonic washing cycles. This enables efficient and controlled separation of both high and low-density oil/water systems with separation efficiency exceeding 98.5%. Reversible wettability conversion is reliably achieved through alternating acid/base treatments. This cost-effective and robust material holds considerable promise for intelligent, controllable oil/water separation applications.

CRedit authorship contribution statement

Chenyu Cai: Project administration, Methodology. **Shuangshuang Liang:** Data curation, Conceptualization. **Tianxiang Li:** Supervision, Software. **Luyao Cong:** Visualization, Validation. **An Fan:** Formal analysis. **Hong Hao:** Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was financially funded by Key Research and Development Program of Shaanxi Province ([2024GX-YBXM-416](#)).

Data availability

No data was used for the research described in the article.

References

- [1] B. Dubansky, A. Whitehead, J.T. Miller, C.D. Rice, F. Galvez
Multitissue molecular, genomic, and developmental effects of the deepwater horizon oil spill on resident gulf killifish (*Fundulus grandis*)
Environ. Sci. Technol., 47 (2013), pp. 5074-5082
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [2] D.I. Little, S.R.J. Sheppard, D. Hulme
A perspective on oil spills: what we should have learned about global warming
Ocean. Coast. Manage., 202 (2021), Article 105509
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [3] C. Chen, D. Weng, A. Mahmood, S. Chen, J. Wang
Separation mechanism and construction of surfaces with special wettability for oil/water separation
ACS Appl. Mater. Interfaces, 11 (2019), pp. 11006-11027
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [4] J. Chen, P. Liu, Z. Guo
Robust special wettability materials for oil-water separation: mechanisms and strategies

Adv. Colloid. Interfac, 335 (2025), Article 103355



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [5] T.-T. Li, S. Li, F. Sun, B.-C. Shiu, H.-T. Ren, C.-W. Lou, J.-H. Lin
pH-responsive nonwoven fabric with reversibly wettability for
controllable oil-water separation and heavy metal removal

Environ. Res., 215 (2022), Article 114355



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [6] T. Yan, T. Zhang, G. Zhao, C. Zhang, C. Li, F. Jiao
Magnetic textile with pH-responsive wettability for controllable oil/water
separation

Colloid. Surface. A, 575 (2019), pp. 155-165



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [7] X. Zeng, K. Yang, C. Huang, K. Yang, S. Xu, L. Wang, P. Pi, X. Wen
Novel pH-responsive smart fabric: from switchable wettability to
controllable on-demand oil/water separation

ACS Sustain. Chem. Eng., 7 (2019), pp. 368-376

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [8] Z. Dang, L. Liu, Y. Li, Y. Xiang, G. Guo
In situ and ex situ pH-responsive coatings with switchable wettability for
controllable oil/water separation

ACS Appl. Mater. Interfaces, 8 (2016), pp. 31281-31288

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [9] H. Gao, Y. Liu, G. Wang, S. Li, Z. Han, L. Ren
Switchable wettability surface with chemical stability and antifouling
properties for controllable oil–water separation

Langmuir, 35 (2019), pp. 4498-4508

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [10] G. Zhu, X. Li, X. Zhang
Fabrication of a superhydrophobic-superoleophilic particle material for oil-water separation and oil extraction
Colloid. Surface. A, 681 (2024), Article 132811
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [11] B. Xiang, Q. Sun, Q. Zhong, P. Mu, J. Li
Current research situation and future prospect of superwetting smart oil/water separation materials
J. Mater. Chem. A, 10 (2022), pp. 20190-20217
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [12] G. Zhu, X. Zhang, Y. He
Facile preparation of superhydrophilic and superoleophobic sand for efficient oil-water separation
J. Water Process. Eng., 61 (2024), Article 105355
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [13] G. Liu, J. Fu, J. Yu, Y. Zhang, R. Wu
Fabricated pH-responsive sponge with switchable wettability for multitasking and controllable oil-water separation
J. Water Process. Eng., 67 (2024), Article 106160
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [14] Y. Wang, W. Zhao, M. Han, J. Xu, X. Zhou, W. Luu, L. Han, K.C. Tam
Topographical design and thermal-induced organization of interfacial water structure to regulate the wetting state of surfaces
JACS Au., 2 (2022), pp. 1989-2000
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [15] B. Qi, B. Fan, B. Xu, M. Zhou, Y. Yu, L. Cui, Q. Wang, P. Wang

Enzymatic construction of temperature-responsive PDMAPS-decorated textiles for oil-water separation

Colloid. Surf. A, 656 (2023), Article 130340



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [16] M.R.B. Domalanta, M.R.R. Ali, R.D. Maalihan, E.B. Caldon
Photo- and thermoresponsive reversible wettability and corrosion protection of electrophoretically deposited TiO₂/pectin composite coatings

Langmuir, 41 (2025), pp. 11958-11967

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [17] J. Yong, F. Chen, M. Li, Q. Yang, Y. Fang, J. Huo, X. Hou
Remarkably simple achievement of superhydrophobicity, superhydrophilicity, underwater superoleophobicity, underwater superoleophilicity, underwater superaerophobicity, and underwater superaerophilicity on femtosecond laser ablated PDMS surfaces

J. Mater. Chem. A, 5 (2017), pp. 25249-25257

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [18] H. Zhai, R. Qu, X. Li, Y. Liu, Y. Wei, L. Feng
Crown ether modified membranes for Na⁺-responsive controllable emulsion separation suitable for hypersaline environments

J. Mater. Chem. A, 8 (2020), pp. 2684-2690

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [19] L. Liu, Y. Pan, K. Jiang, X. Zhao
On-demand oil/water separation enabled by magnetic superoleophobic/super-hydrophilic surfaces with solvent-responsive wettability transition

Appl. Surf. Sci., 533 (2020), Article 147092

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [20] Q. Gao, J. Zhao, J. Hu, M. Wang
Applying a switchable superhydrophobic and hydrophilic ZnO nanorod array-coated stainless-steel mesh to electrically-induced oil/water separation

Colloid. Surf. A, 628 (2021), Article 127231

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [21] K.-Y. Kim, R.P. Srivastava, D.-Y. Khang
Oleophilic to oleophobic wettability switching of isoporous through-hole membranes by surface structure control for low-voltage electrowetting-based oil-water separation

J. Membr. Sci., 646 (2022), Article 120281

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [22] H. Zhang, C. Zhao, Y. Yang, Y. Sheng, S. Peng
Fluorine-free fabrication of a magnetic-responsive sponge as absorbent for highly efficient and ultrafast oil-water separation

Thin Solid Films, 751 (2022), Article 139205

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [23] X. Chang, Z. Wang, S. Quan, Y. Xu, Z. Jiang, L. Shao
Exploring the synergetic effects of graphene oxide (GO) and polyvinylpyrrolidone (PVP) on poly(vinylidene fluoride) (PVDF) ultrafiltration membrane performance

Appl. Surf. Sci., 316 (2014), pp. 537-548

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [24] F. Yang, Z. Guo

A facile approach to transform stainless steel mesh into pH-responsive smart material

RSC Adv., 5 (2015), pp. 13635-13642

[View in Scopus ↗](#) [Google Scholar ↗](#)

[25] H.H. Liu, W.K. Chin

Surface morphologies and wettability studies of pH-sensitive nanofibrillated membranes

ChemistrySelect, 8 (2023), Article e202301391

[View in Scopus ↗](#) [Google Scholar ↗](#)

[View Abstract](#)

Recommended articles

Based on reading popularity

Research article Full text access

One-stone-two-birds design: ZIF-8@OCNTs/PVDF mixed matrix membranes with dual-functionality for complex oily wastewater remediation

Yao Chen, ..., Yuanju Li

Journal of Environmental Chemical Engineering • Volume 13, Issue 5 • 2025 • Article 118673



[View PDF](#)

Research article Full text access

Constructing fluorine-free and photothermal MXene/candle soots-based superhydrophobic sponge for oil/water separation, crude oil recovery, and anti-icing

Mingkun Wang, ..., Zhiwei He

Process Safety and Environmental Protection • Volume 202, Part A • 2025 • Article 107738

 [View PDF](#)

Research article Open access

Improved electrochemical sensor using functionalized silica nanoparticles (SiO₂-APTES) for high selectivity detection of lead ions

Afef Dhaffouli, ..., Houcine Barhoumi

Materials Chemistry and Physics • Volume 318 • 2024 • Article 129253 [View PDF](#)

Research article Full text access

Sesame pods derived high-efficiency demulsifier: Synthesis and demulsification performance in oily wastewater treatment

Dou Wang, ..., Qingwen Qin

Process Safety and Environmental Protection • Volume 203, Part B • 2025 • Article 107976 [View PDF](#)

Research article Full text access

Biomimetic slippery-hydrophilic patterned membrane for efficient emulsion separation

Guangyue Du, ..., Yuzhong Zhang

Journal of Environmental Chemical Engineering • Volume 14, Issue 2 • 2026 • Article 121645 [View PDF](#)

Research article Open access

A novel homogeneous amphipathic PTFE/ β -FeOOH nanofiber membrane for emulsion separation and photocatalytic degradation in harsh environments

Fan Liu, ..., Qinglin Huang

Advanced Membranes • Volume 6 • 2026 • Article 100187 [View PDF](#)

Cited by (1)

Research article

All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.

Citation Indexes

1

Captures

Mendeley Readers

1

[View details ↗](#)

© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.